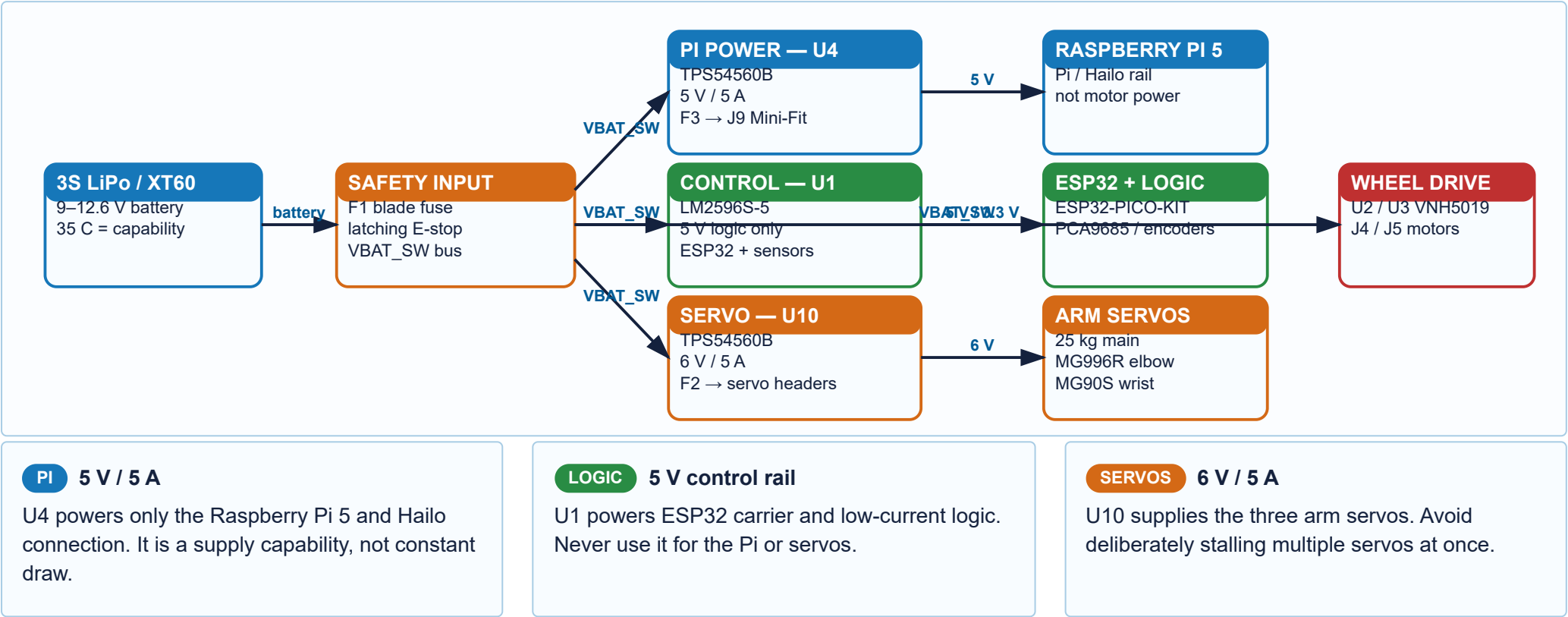


Pickleball Robot Controller

Readable wiring and build book · Rev R1 · companion to the KiCad PCB source

Use this PDF to understand and wire the robot. The KiCad schematic remains the authoritative PCB/netlist file; this version deliberately splits the system into readable pages.

1. Power architecture



2. Raspberry Pi, ESP32, and control wiring

Pi ↔ ESP32 UART: J10

J10 signal	Connects to	Purpose
Pi TX	ESP32 RX0 / GPIO3	Pi sends drive commands
ESP32 TX0 / GPIO1	Pi RX	ESP32 returns status
GND	Pi GND	Required shared reference
I²C SDA/SCL	GPIO21 / GPIO22	Optional expansion / servo controller

UART is 3.3 V only. Do not connect a 5 V UART signal. Do not actively use Pi UART while the ESP32 USB serial adapter is driving RX0/TX0.

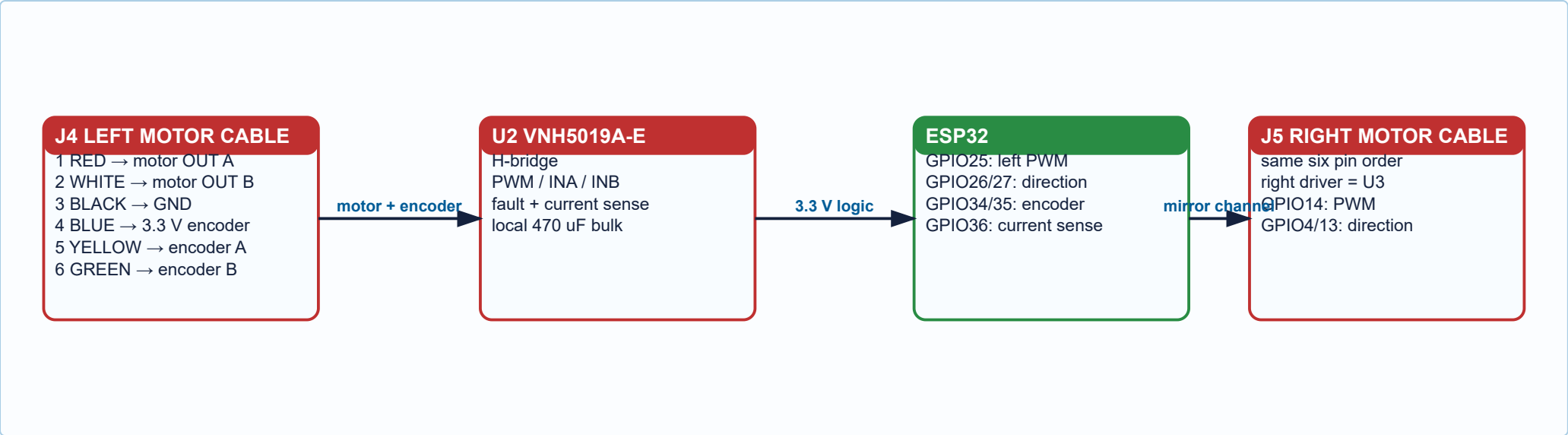
ESP32 socket assignments

Function	ESP32 pin
Left PWM / direction	GPIO25 / GPIO26 / GPIO27
Right PWM / direction	GPIO14 / GPIO13 / GPIO4
Left / right encoder	GPIO34/35 and GPIO37/38
Battery ADC	GPIO33
Servo I²C	GPIO21 SDA / GPIO22 SCL

Board status LEDs

GREEN logic 5 V **BLUE** Pi 5 V **AMBER** servo 6 V **RED** motor fault

3. Wheel motors and encoders

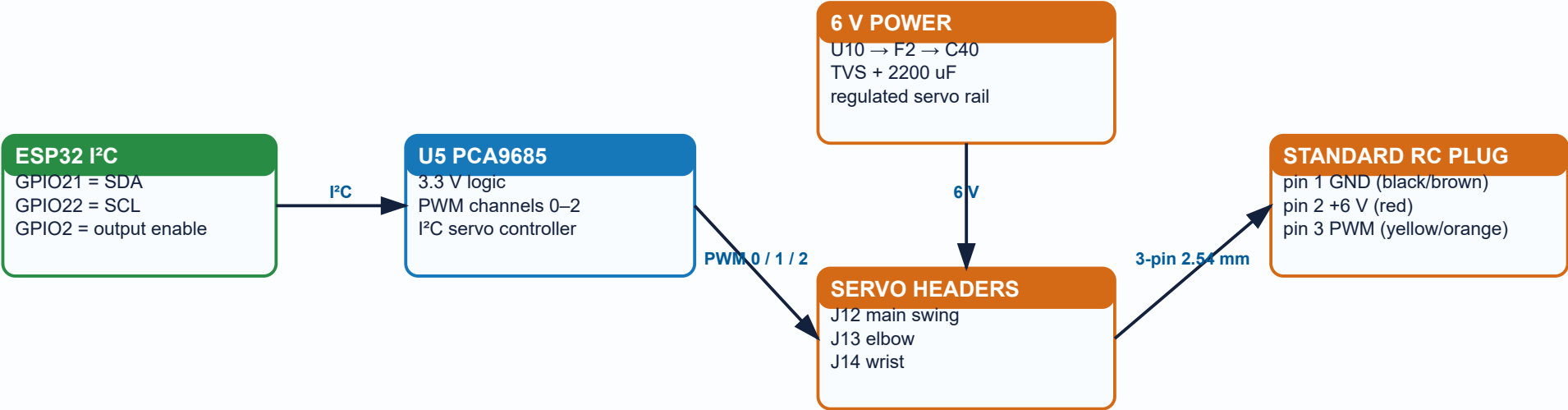


Both six-pin motor connectors: J4 left and J5 right

Pin	Cable colour	Function	Important check
1	Red	Motor terminal A	If a wheel goes backwards, swap the two motor wires in software or at the connector.
2	White	Motor terminal B	
3	Black	Encoder ground	Confirm this exact order against the plug before ordering the PCB.
4	Blue	Encoder +3.3 V	
5	Yellow	Encoder channel A	
6	Green	Encoder channel B	

Connector: 6-pin JST-PH, 2.00 mm pitch. The board footprint assumes the 10 mm pin-1-to-pin-6 measurement you made.

4. Three-degree-of-freedom arm servos



What plugs where

Header	Servo	PCA9685 channel
J12	Main 25 kg swing servo	0
J13	MG996R elbow servo	1
J14	MG90S wrist servo	2

Safe first test

1. Power the board from a current-limited bench supply.
2. Check the amber LED and measure 6.0 V at a servo header.
3. Connect one servo only, with arm free to move.
4. Use slow movements and PWM limits before connecting all three.

5. Build and PCB-layout checklist

Easy normal-iron parts

- All 0805 resistors and small ceramic capacitors
- Through-hole 220 μ F / 1000 μ F / 2200 μ F electrolytic capacitors
- JST-PH motor headers, Pi connector, XT60 pigtail connector, servo headers
- Fuses and large terminal parts

Heat gun / paste parts

- U2/U3 VNH5019 motor drivers
- U4/U10 TPS54560B buck regulators
- U5 PCA9685 and large power inductors

The exact manufacturer part numbers are in `ORDERING_GUIDE.md`.

Do before ordering

1. Open the KiCad project and assign/verify every footprint.
2. Place the high-current parts first; use wide copper pours and thermal vias.
3. Keep buck switch nodes short and away from encoder/UART traces.
4. Keep ESP32 antenna area clear of copper and components.
5. Run ERC, then update PCB from schematic and run DRC.
6. Print J4/J5 at 1:1 scale and compare with your physical motor plug.

Do not power the Pi or servos from the ESP32 logic rail. Each has its own regulator on the same 3S battery.